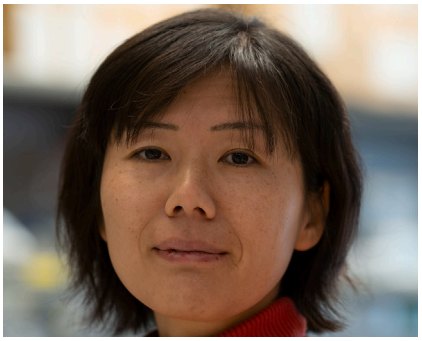
Developmental Cell

Voices

Understanding how cells and organisms keep time during development

A classical question in biology is how different processes are controlled in space and time, with research pointing to different mechanisms as timers. In this collection of Voices, we asked researchers to define their scientific questions related to time-keeping and the approaches they use to answer them.



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Photo credit: Humboldt Foundation

Quest for universal tempo modulators

While the pace of embryonic development is robust, it varies significantly across species. For example, the gestation periods of mice, humans, and elephants are 20 days, 9 months, and 22 months, respectively. Despite these widely recognized empirical differences, the molecular mechanisms of species-specific developmental tempo remained largely unexplored until recently due to the absence of suitable experimental models. With the advancement in stem cell-derived models, we can now recapitulate *in vitro* the tempo of several developmental processes of traditionally inaccessible species, such as humans and even rhinoceroses. Cross-species comparison of stem cell models has identified a couple of candidates for the modulators of developmental tempo: species-specific biochemical reaction rates, metabolic factors, and epigenetic regulations, among others.

However, these tempo modulators have been studied for limited species, genes, and developmental processes. We do not know, for example, if a single metabolic factor can explain the tempo of multiple species, if reaction rates differ between species for all or specific genes, and if epigenetic regulation is equally important for different processes, such as the neurogenesis and segmentation clock. Addressing the universality of tempo modulators and clarifying their epistatic relationships are crucial remaining challenges. Ultimately, the modulators of species-specific developmental tempo should be encoded in the species genomes. The stem cell zoo should meet comparative genomics and parallel genome engineering in the future.



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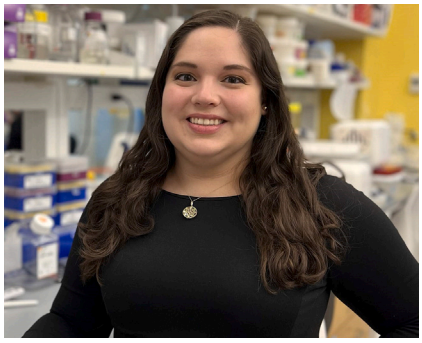
Cracking the molecular tick-tock code

The temporal control of gene expression guides how an organism is built in development. However, the reasons behind species-specific rates of development remain unclear. My research aims to understand how the rate of development is measured and controlled autonomously in cells. We do so using comparative stem cell models and studying natural tempo modulation in embryogenesis.

In my previous work, we found that the differences in tempo were not due to differences in the sensitivity of the cells to signals, nor in the sequence of *cis*-regulatory elements for key genes. Instead, we made the exciting discovery that mouse proteins are degraded twice as fast as human and the cell cycle is twice as long, mirroring differences in developmental tempo. Building on this work, we are now using dynamic stable isotobe labeling by amino acids in cell culture (SILAC) to identify which specific proteins do and do not follow this general fold change in mouse versus human neural progenitors. We hypothesize that specific very short- or long-lived proteins may act as a rate-limiting step for the rate of protein degradation and affect tempo.

We also want to decipher how tempo can be modulated, and we are using diapause as a model to perturb tempo. Mouse embryos slow down development for a few days after diapause induction and become quiescent thereafter. Our goal is to determine whether the mechanisms employed during the onset of diapause are analogous to those tempo mechanisms identified in other contexts. We are focusing on the role of the speed of biochemical reactions and epigenetics to unravel generalized principles of developmental tempo.

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Margarete Diaz-Cuadros
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The fourth dimension

Despite broad conservation of developmental programs, the speed of development is highly species specific. For instance, mice complete embryogenesis in 15 days whereas elephants do so in 116 days. In the last few years, stem cell-based models of mammalian development have ushered in a renaissance in the study of developmental tempo. Several mechanisms have been proposed to regulate developmental speed, from epigenetics to biochemical reactions speeds, and mitochondrial metabolism. The overarching conclusion from these studies is that the speed of development is species specific because the most basic molecular processes within the cell progress at distinct rates. It is profoundly surprising that such deeply conserved processes, which are normally considered house-keeping functions, differ in speed even among closely related organisms.

Going forward, it will be exciting to uncover how the variety of mechanisms proposed to control developmental speed fit together into a regulatory timing pathway. As changes in developmental speed have occurred independently along multiple lineages, it will also be fascinating to detangle the diverse ways in which evolution has modulated developmental speed. This will require researchers to look beyond mammals and embrace unbiased approaches. Equally important will be to identify how developmental speed is encoded in the genome. Advances in this field will make possible the manipulation of developmental speed for translational applications, yet these advances will need to be accompanied by careful ethical considerations.



Kevin J. Chalut
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On ruffles and noise in cell sorting

It was meant to be a low-lying fruit. Previously it had been shown that binary cell sorting could be driven by differences in interfacial cell surface tension. This should explain why epiblast (EPI) and primitive endoderm (PrE)—early mouse embryo lineages—would sort. But upon doing relevant cell measurements, those differences were not there: no differences in surface tension, or any other static mechanical property, could explain their sorting. So we set the project aside.

One day, we were looking at old micrographs of early embryo cells and Jenny Nichols, who drove this project with me, noted they used to distinguish PrE cells by their greater "ruffliness". I wondered whether that ruffliness might be blebs, a signature of dynamic instabilities in cell surface mechanics. Indeed, we found the PrE cells were far more blebby than EPI cells. Then, we changed our view—maybe the cell sorting was not happening because of static differences like surface tension, but instead because of distinct dynamic properties, like surface fluctuations. We went on to show with a mix of modeling and experiments that the surface fluctuations led to a greater fluidity in the PrE cells and were indeed a mechanism of sorting. This sorting mechanism will be a general feature of many binary mixtures of cells in development and disease, especially in the context of signaling affecting cell mechanics. Ultimately, we underlined the emerging notion that for any biological problem, we need to think not just of what cells look like in an instant, but also how they behave dynamically over time.



Gang Wu
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Timing of juvenile-to-adult transition in plants

Post-embryonic plant development progresses through juvenile, adult, and reproductive phases. The transition from juvenile to adult phase is known as vegetative phase change, while the progression from adult to reproductive phase is termed floral induction. The mechanism behind achieving coordination between these distinct phases poses a central challenge in plant developmental biology.

In contrast to floral induction, the mechanism of vegetative phase change remained largely obscure until 2006 when my work revealed that vegetative phase change is regulated by the temporal decline in miR156 expression and the corresponding increase in *SPL* expression. Epigenetic factors synchronize the temporal reduction in miR156, primarily by balancing H3K27ac and H3K27me3 at miR156 loci. Despite progress in comprehending the role of histone modifications at miR156 loci in its temporal regulation, the mechanism behind the timing of H3K27me3 deposition at these loci still awaits further investigation.

Phenotype-based genetic analysis has identified numerous miR156 regulators, impacting its overall level rather than its temporal expression. Leveraging the whole genome sequences of over 1,000 *Arabidopsis* natural accessions, the utilization of these accessions alongside GWAS analysis has facilitated the identification of novel regulators influencing miR156 expression. This approach will continue to unveil new regulators of miR156 that impact its temporal expression, offering fresh insights into the molecular mechanisms governing the timing of the juvenile-to-adult transition in plants.



Antony N. Dodd
John Innes Centre

Circadian timing in plants and their neighbors

Living organisms experience 24-h changes in their environments due to the rotation of the Earth on its axis. This challenges plants, as they require light for photosynthesis, daily temperature fluctuations impact their metabolism and physiology, and there are daily behaviors of pathogens and pollinators. The selection pressures arising have led to the evolution of circadian clocks. Circadian clocks produce a biological estimate of the time of day that, in plants, control activities including growth and developmental transitions. Circadian rhythms contribute to the fitness of plants. *Arabidopsis thaliana* is a powerful model to study circadian rhythms, with an excellent range of experimental tools to support both discovery science and translation into crops.

There are many exciting open questions in the field. One concerns the effect of climate change upon the contribution of plant circadian clocks to the transition from vegetative to reproductive development. Given that plant reproductive development is essential for our food supply, this question could be of profound socioeconomic importance. A further exciting area represents the contribution of plant circadian programs to ecological interactions. Plants develop close associations with microbial communities, which improve soil nutrient availability and pathogen protection. The formation of and interaction between circadian programs in plants and their interacting microbial communities represents a major emerging topic, particularly since the recent discovery of circadian clocks in non-photosynthetic bacteria.



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Keeping track of time at the beginning of life

The concept of time in development is intrinsic to the progression of cellular states. Molecularly, it ranges in scale from seconds (transcription) to years (human oocytes arrested in meiosis).

After fertilization, chromatin organization is drastically remodeled to restore totipotency and enable a new developmental program. The radial positioning of the genome in the nuclear 3D space with the formation of lamina-associated domains precedes chromosome folding into A- and B-compartments. Recently, the single-cell investigation of replication timing established that the replication program in embryos is initially not well defined when development begins. DNA replication is slow and fork speed increases as development proceeds. Perturbations leading to a decrease in fork speed promote reprogramming, underlying the role of DNA replication and perhaps also its speed for cell fate changes.

In the future it will be essential to identify causal relationships—if any—and to establish how the molecular layers of the epigenome come together to support cellular plasticity. Can we modulate one of these layers and affect another? Or are they robust to perturbations of the other epigenome layers? Identifying the regulators of epigenome remodeling is necessary to address these questions. Adapted small-scale screenings, using low-input genomic approaches, will be fundamental to uncovering molecular regulation. Studies in embryos have indeed taught us that the relationships established in cultured cells do not always hold in developmental settings.



Michael Levine
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Enhancer autonomy and nonautonomy in development

Classical studies of developmental enhancers relied on the use of fixed embryos and therefore focused on spatial limits rather than temporal dynamics of gene regulation. Recent live imaging methods provide insights into timing mechanisms. Synthetic reporter genes containing minimal enhancers often recapitulate normal spatial limits, as seen for eve stripe 2, but fail to produce authentic temporal dynamics. Developmental enhancers therefore appear to be autonomous with respect to spatial limits but nonautonomous for correct temporal dynamics. What features of endogenous chromosomal loci influence timing?

Recent studies suggest several potential influences, including 3D genome organization and noncoding RNAs. Both factors may contribute to the timing of the Scr Hox stripe of expression in the early *Drosophila* embryo. Timely activation depends on interactions between a pair of tethering elements that map near the Scr promoter and distal enhancer. Disruption of this regulatory loop causes a delay in expression and a weak homeotic phenotype. A noncoding RNA emanating from the Scr embryonic enhancer might also influence the timing of Scr activation.

MS2/PP7 live-imaging methods have revealed new insights into developmental timing mechanisms. An even deeper understanding would be obtained by extending these methods to the visualization of 3D chromosomal dynamics in living embryos.



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Unlocking the clock: Biomarkers of aging

The recent development of quantitative biomarkers of aging has revolutionized the field. The first integrative aging biomarkers (aging clocks) were developed about 10 years ago. They were based on DNA methylation changes and trained on chronological age. Later, other omics modalities were used, with clocks trained on phenotypic changes, future mortality, and aging pace. These tools allowed researchers to address many important questions in aging biology. For example, aging clocks trained for adult ages were found to tick already during development, challenging the idea that aging begins after the onset of reproduction. In fact, clocks begin to tick around gastrulation, marking the "ground zero" of aging. Interestingly, reprogramming adult cells to induced pluripotent stem cells reduces biological age close to zero, and rejuvenation also occurs upon heterochronic parabiosis in mice and during early embryonic development.

However, challenges remain. Some researchers view aging clocks as the glass half empty, while others see them as half full. Clocks that quantify aging progression do not necessarily measure the effects of anti-aging interventions, and it remains unclear

how they should be constructed and what they should quantify. This challenge partly relates to differing views on aging among scientists. Recently, the first causal epigenetic clocks were reported, and it was shown that aging clocks have both stochastic and regulatory components. The excitement about these biomarkers is that they may be able to assess interventions, without waiting for age-related diseases or mortality.

DECLARATION OF INTERESTS

M.E. is a visiting group leader in the European Molecular Biology Laboratory (EMBL) Barcelona. M.-E.T.-P. is a member of the Ethics Advisory Panel of MERCK. M.-E.T.-P. is full professor of the Ludwigs Maximilians University in Munich.